

Vaidik Shah

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Education

California State University, Long Beach

Master of Science in Computer Science

Key Courses: Artificial Intelligence, Machine Learning, Distributed Computing, Natural Language Processing

May 2027

Long Beach, CA, USA

Gujarat Technological University

Bachelor of Science in Computer Science

Key Courses: Object Oriented Programming, Data Structures, Relational Database, Computer Networks, System Design

June 2024

Ahmedabad, India

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, C++, C#, Golang, Rust

Backend & Databases: Node.js, Express.js, SQL (MySQL, PostgreSQL), NoSQL (MongoDB, DynamoDB)

AI/ML: PyTorch, TensorFlow, Scikit-learn, XGBoost, R, LangChain, NLP, Computer vision, NumPy, Pandas, Tableau, SAS

Cloud/DevOps & Web: AWS, GCP, Azure, CI/CD, Git, Linux/Unix, React, Angular, Vue, Bootstrap, Web3, HTML5/CSS

Software Engineering Fundamentals: Algorithms, Debugging, Scalable Systems, SDLC, Multi-threaded Programming, Agile

Experience

Technolee

July 2024 - July 2025

Software Engineer (Data & ML)

Ahmedabad, India

- Engineered and deployed an end-to-end REST API that delivered **ML predictions < 200ms**, enabling **real-time dashboards for 20+ product lines for cross-functional teams**, eliminating **10+ hours of manual reporting each week**.
- Built and deployed a **logistic regression model on 30K+ sales records**, applying **machine learning and statistical modeling techniques** to improve **demand forecasting accuracy by 25% across 20+ product categories**.
- Automated **PostgreSQL + NLP ETL pipeline** with custom scheduling, **deployed on AWS**, processing **50K+ daily records** and enabling analysts to work with daily **refreshed insights instead of weekly updates**.

Indian Space Research Organization (ISRO)

January 2024 - July 2024

AI/ML Engineer Intern

Ahmedabad, India

- Built **enterprise satellite data pipeline** with automated validation protocols, **achieving 99.9% data integrity**, while processing 1+GB of datasets daily for solar insolation forecasting in renewable energy applications.
- Optimized **LSTM-based forecasting models using PyTorch/CUDA and MATLAB**, leveraging **advanced statistical techniques** to **reduce prediction error by 25%** and **improve model efficiency by 20%**.
- Developed **data visualization dashboards using Matplotlib and Seaborn**, transforming complex satellite data into actionable insights and **reducing reporting cycles by 40%**.

Cre-Art Solutions

June 2023 - August 2023

Software Engineer Intern

Ahmedabad, India

- Redesigned Django REST APIs** with advanced OOP patterns, achieving a **20% performance boost** while processing **50K+ daily financial transactions** for a trading platform.
- Built an **automated validation framework** with **real-time error detection**, reducing **accounting discrepancies by 18%** and ensuring financial compliance standards across high-volume transaction processing.
- Built interactive **data visualization dashboards in Power BI with secure SQL integration**, enabling **real-time financial analysis** and **reducing monthly reporting time by 30%**.

Personal Projects

PageSpeaks — Multi-Voice TTS Platform | [Try me!](#) | Python, PyTorch, F5-TTS, Whisper, Modal, FastAPI, Next.js

May 2026

- Fine-tuned **F5-TTS** (flow-matching voice synthesis) on **19 hours** of lecture audio, producing **6,676 training clips** and deploying on **serverless Modal GPU** with **HuggingFace Hub** model storage.
- Built an **end-to-end audio preprocessing pipeline in Python** — **silence-splitting**, **noise reduction**, and **Whisper-based transcription** — converting raw MP3 lectures into a clean **LJSpeech-format** training dataset.
- Implemented **chunked streaming inference on serverless GPU**, **prefetching** the next audio segment while the current one plays to minimize perceived latency across long-form text.

AI-Powered Professor Rating Platform | [Try me!](#) | LangChain, FastAPI, Redis, WebSockets, JavaScript

August 2025

- Built an AI-powered professor search platform using **LangChain + Pinecone (RAG)** on RateMyProfessor data, enabling students to discover and evaluate professors with significantly improved relevance over traditional keyword search.
- Designed a **FastAPI WebSocket** backend with **Redis caching** that streams token-by-token responses in real time, with an agentic RAG-to-GraphQL fallback pipeline ensuring consistent answer coverage across queries.
- Developed a responsive web interface with HTML/CSS featuring **AI-driven professor recommendations**, follow-up context handling, and side-by-side professor comparisons for an interactive search experience.

SCertification

AWS Certified Solutions Architect – Associate (Credential ID: [link](#))

Expires March 2029